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# Digital connectivity and mental health symptoms among women in Bangladesh: a rural–urban perspective

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## Abstract

**Objective** To examine the associations between internet use, information and communication technology (ICT) usage, and ICT ownership with symptoms of depression and anxiety among ever-married women in Bangladesh, with particular attention to rural–urban disparities and spatial inequities.

**Methods** This study analyzed nationally representative cross-sectional data from 18,987 ever-married women aged 15–49 years collected from the Bangladesh Demographic and Health Survey 2022 (BDHS). Symptoms of depression and anxiety were assessed using the Patient Health Questionnaire−9 (PHQ-9) and the Generalized Anxiety Disorder-7 (GAD-7), respectively. Survey-weighted multivariable logistic regression models were fitted to estimate associations between digital connectivity and mental health symptoms, with descriptive stratified analyses by place of residence (rural/urban).

**Results** Among all women, 4.8% women showed depression symptoms, while 4.1% showed anxiety symptoms. Among rural women, ICT use was significantly associated with higher odds of depression symptoms (aOR:1.38; 95% CI 1.12–1.69). In urban settings, higher ICT ownership was associated with both depression symptoms (aOR:1.81; 95% CI 1.01–3.25) and anxiety symptoms (aOR:2.11; 95% CI 1.14–3.92). Rangpur and Rajshahi division showed lower digital access and distinct mental health patterns.

**Conclusions** Digital connectivity and women's mental health symptoms were associated across rural–urban and geographic contexts in Bangladesh. The observed disparities in anxiety symptoms and depression symptoms across various dimensions of digital access indicate the importance of considering digital access within broader negotiations of women's health symptoms. Combining mental health symptoms progress with gender-sensitive and geographically targeted digital strategies is important for progressing women's mental well-being and achieving Sustainable Development Goal (SDGs)3.


## Highlights

- High mental health burden was highlighted in digitally connected divisions Dhaka, Chattogram, Sylhet, and Barishal.
- Low digital access and lower mental health burden were concentrated in Rangpur.
- These findings reveal a significant rural–urban and geographic inequities, emphasizing the need for region-specific, equity-oriented digital mental health interventions in Bangladesh.

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**Keywords** Digital connectivity · Mental health inequities · Anxiety and depression · Women's health · Rural–urban disparities · Spatial inequality · ICT exposure · Bangladesh

**Abbreviations**

| | |
|---|---|
| BDHS | Bangladesh Demographic and Health Survey |
| WHO | World Health Organization |
| GAD-7 | Generalized Anxiety Disorder-7 |
| PHQ-9 | Patients Health Questionnaire−9 |
| IPV | Intimate Partner Violence |
| VIF | Variance Inflation Factor |
| MDG | Millenium Development Goal |
| CAPI | Computer assisted personal interview |
| SDG | Sustainable Development Goal |
| AOR | Adjusted Odds Ratio |
| CI | Confidence Interval |

## 1 Introduction

### 1.1 Background

Mental health symptom is a major cause of disability worldwide, affecting over 970 million people, approximately one in eight individuals globally [1]. Specially, for reproductive aged women, mental health symptoms involve extremely to the global burden of disease and are associated with substantial social, economic, and health consequences [2].

ICT ownership refers to the availability of communication devices in a household, such as televisions, radios, mobile phones, and smartphones [3, 4]. It indicates basic access to information and communication technologies and is often used as a proxy for household access to information services [5]. ICT usage refers to individuals' actual connection with media content through these devices [6]. It is generally evaluated by self-reported frequency of exposure, such as reading newspapers, listening to the radio, or watching television, particularly in settings with limited detailed usage data [7, 8]. In this study, digital connectivity is regarded a comprehensive concept, including access to communication technologies, individual internet engagement, and exposure to information sources [9]. Due to each features indicates a distinct characteristic of digital connectivity; these components were evaluated separately [4].

In South Asia, the burden of mental health symptoms from socioeconomic inequalities, cultural stigma, and limited healthcare infrastructures [10, 11]. Rapid demographic and technical evolutions in Bangladesh restructured social conditions; however, the mental health symptoms of reproductive aged women remain limited explored and receives limited policy attention [12]. Bangladesh, a lower-middle-income country (LMICs) with more than 165 million people, bears an underestimated and significant burden of mental symptoms among women, particularly among ever-married women who are suffering to experience justification attitude towards violence (IPV), including reproductive health issues, and limited decision-making autonomy [13, 14]. The available evidence from Asia and Africa indicates heterogeneous depression and anxiety prevalence among women, with risk factors including age, education, parity, internet use, reproductive coercion, and empowerment status [3, 15, 16].

The BDHS interviews only ever-married women aged 15–49 years at the national level and first time included mental health screening tools of PHQ-9 and GAD-7 [17]. Additionally, this survey is mainly collected the reproductive, maternal, and child health monitoring interest, which is most relevant to married women in the Bangladeshi context [17]. In Bangladesh, recent cross-sectional studies showed moderate to high prevalence of anxiety

and depression among women, associated with sociodemographic, reproductive, and decision-making autonomy [2, 18, 19]. Besides, novel data of mental health symptoms were added in BDHS 2022 with limited exploration about how digital inclusion in the form of internet access, ICT ownership, and use is associated with mental health symptoms [20].

The primary causes of poor mental health symptoms among women of reproductive age include pregnancy loss, IPV, limited decision-making autonomy among women, poor nutrition, and socioeconomic status [21, 22]. However, several studies showed that some media access factors particularly household internet status, ICT ownership and usage also play a vital role in mental health [23–25]. Excessive internet usage was associated with mental health symptoms, lifestyle disruptions, and reduced well-being [26, 27]. One study revealed that optimal levels of internet use and ICT ownership were significantly associated with better mental health outcomes, and another study in Bangladesh showed ICT ownership, and ICT usage was significantly associated with lower anxiety, and depression [28]. Rural women face limited health information and knowledge gaps compared to urban women in Bangladesh, due to limited facilities of digital access [29]. Although several studies have explored individual determinants of mental health status among ever married women, limited research has systematically examined the association between mental health status, particularly anxiety and depression and internet usages, ICT ownership, and ICT usage using large national survey data, geospatial, and rural–urban stratified analysis in the Bangladeshi context [30].

This study provides significant findings through its multidimensional analysis of digital connectivity through internet usage, ICT access, and ICT usages regarding anxiety and depression among Bangladeshi women based on nationally representative data from BDHS 2022 [31]. The study also helps with to the existing literature by considering geographical differences between urban and rural settings [32]. Thus, the association between mental health symptoms and internet use, ICT exposure among women remains underexplored [33]. Understanding how digital access (internet use, ICT usages, and ICTs ownership) associates with mental health symptoms can inform equity-focused interventions to progress SDGs goal 3 (mental health and well-being) [34].

Therefore, the primary objective of this study is to assess the association of mental health symptoms (anxiety and depression) with internet use, ICT usages, and ICTs ownership with rural–urban inequalities. The additional objective is to assess administrative division-wise descriptive geospatial inequalities of the mental health symptoms, internet use, ICT usages, and ICTs ownership.

## 2 Methodology

### 2.1 Data source, study design and settings

This study is based on the cross-sectional secondary data set of the most recent 2022 BDHS survey, which was the 9th round nationwide cross-sectional survey [17]. This survey was conducted by the National Institute of Population Research and Training (NIPORT) in collaboration with the Health Education and Family Welfare Division of the Ministry of Health and Family Welfare, Government of the People's Republic of Bangladesh. The survey was implemented by Mitra and Associates, and ICF International provided technical assistance. The survey was conducted in four phases, each lasting approximately 4 weeks, from June to December 2022 [17].

### 2.2 Sample size, and sampling technique

The BDHS 2022 employed a two-stage stratified cluster sampling procedure. In the first stage, a probability proportional to size approach was used to select 674 EAs. The second stage involved selecting 30,330 households from the selected enumeration areas (EAs). In this study, 30,358 ever-married women aged 15–49 years were eligible to experience individual interviews, where 30,078 participants completed the interview process, giving a response rate of 99.1%. Two different complete case analyses were used in this paper according to data

availability. The sample of women who experienced the analysis regarding ICT ownership and usage had 18,987 women. Observations with missing values for outcome variables (PHQ-9 and GAD-7 scores), exposures, and covariates were excluded. Regarding the internet usage models, the analyses were conducted using a restricted complete-case sample because some of the data about internet use in the BHDS data set was unavailable. As a result, the models concerning internet usage were calculated independently. For internet use analysis, a restricted complete-case sample of 5,486 women was used due to missing internet use information in the BDHS dataset. The BDHS-2022 report discusses, in detail, about sampling process, data collection procedure, quality control, and survey questionnaire (Household questionnaire, Individual questionnaire, etc.) [17]. This study follows the (Strengthening the Reporting of Observational Studies in Epidemiology) STOBE guidelines. No patients are included in this study (Fig. 1).

## 3 Outcome variables

The study considered two primary outcome variables such depression and anxiety. Depression was assessed using PHQ-9, and any score equal to or greater than 10 was regarded as severe depressive symptoms. The variable was also dichotomized to identify 1 as PHQ-9 scores ≥10 and 0 for scores <10. Anxiety was also assessed using the GAD-7 questionnaire, and a score of 10 or more was used to denote clinically relevant anxiety. Anxiety was operationalized as a dichotomous variable (1 for GAD-7 ≥10, otherwise 0), similarly depression. This process is consistent with previous DHS analytical framework [2, 32].




**Fig. 1** Flowchart of selection of women included in the final analysis

### 3.1 Exposures

Three exposure variables were constructed to measure different dimensions of digital access and media consumption**.** Internet use was measured using the self-reported frequency variable and re-coded into a binary measure. Respondents who reported never or rarely (before last 12 months), the internet were labelled as non-users (0), while respondents use last 12 months, were labelled as users (1). Second, ICT ownership was operationalized as a composite of four device ownership indicators: television, radio, mobile phone, and smartphone. Each device was assigned the value 1 if owned and 0 otherwise. A range score of 0 to 4 was developed to indicate total ICT ownership. The same was dichotomized into two categories: no ownership (0) and any ownership (1, for one or more devices). Third, ICT usage assessed frequent exposure to traditional media sources. Respondents' frequency of use of newspapers, radios, and television sets were each recorded into binary indicators, with frequent use (code 2) being recorded to 1 and infrequent or no use (codes 0–1) recorded to 0. The three indicators were added up to create a total ICT usage score ranging from 0 to 3, and dichotomized as: no frequent use of any medium (0) and frequent use of one or more mediums (1). These variables were designed to capture both ownership (access) and usage (use) dimensions of ICT for mental health symptoms.

The independent exposure variables were constructed based on the standardized indicators of the Bangladesh DHS survey of 2022 and the existing literature based on the previous DHS surveys exploring the ICT ownership and media exposure as well as their impact on the health status of women. ICT ownership denoted the availability of communications technology at household level while internet use indicated personal level of connection [23, 35].

### 3.2 Covariates

A number of covariates at the individual level were included to adjust for any potential confounding between internet use and mental health symptoms. Sociodemographic characteristics covered were maternal age, education, religion, place of residence, administrative division, household wealth, household size, sex of the household head, current work status, and father's education. Age was classified into seven classes: 15–19, 20–24, 25–29, 30–34, 35–39, 40–44, and 45–49 years. Women's education was classified as no or primary education, secondary incomplete, and secondary or higher. Religion was classified as Muslim or non-Muslim. Residence was classified as urban or rural, and region was indicated by administrative division. The wealth index of the household was re-coded into three categories: poor (poorer and poorest), middle, and rich (richest and richer). Household size was categorized into small (1–3 members) and large (4 members and more). Sex of the head of the household and current working status of the respondent were also controlled. The education level of the father was controlled as a categorical variable. Reproductive characteristics were: number of surviving children, currently pregnant, recent menstruation, not having had sex, marital history, and living with husband or spouse. Surviving children numbered none, 1–2, or 3 or more. Women were classified as currently pregnant if they replied yes to being pregnant at the time of the survey. Menstruation in the past six weeks and self-reported never having had sexual intercourse in the current period were both recorded as binary variables. The marital unions were categorized as once or more than once, and current co-residence with husband/partner was recorded as a binary variable too. Husband's desire for fertility was recorded as desired as many or more children as the woman had fewer, and women were also asked if they ever felt pressured to get pregnant. Decision making autonomy was assessed using five decision-making indicators: health care, purchases in the household, visits from family or relatives, money control, and family planning. Decision-making autonomy over contraceptive use was also coded respondent/joint decision vs. husband/other. Gender attitudes were captured by five justifications of intimate partner violence (IPV) questions. A dummy variable was created to indicate whether the respondent justified IPV for at least one condition. All variables potential covariates for this study used based on DHS based study and BDHS 2022 dataset availability [32, 36–39].

### 3.3 Confounder selection

Basically, the chi- square test conducted to assess the crude association between outcome and exposure then ($p < 0.025$) consider for potential confounder and overall priority based on theoretically and conceptually [40].

Based on previous literature, availability of variables in DHS data, theoretical, and conceptual considerations regarding potential confounders associated with digital connectivity and mental health symptoms selected a priori based on combination of DAG. Primarily, the DAG was used to assess a minimally adequate adjustment set. Additionally, Chi-square tests were conducted only to define crude associations and were not used as the primary measure for covariate selection. Overall, final survey-weighted multivariable logistic regression models included theoretically related confounders acknowledged using the conceptual outline. This model building process is consistent with previous DHS based study [2, 36] (Fig. 2).

### 3.4 Final adjusted models covariates

Based on the conceptual framework, theoretical relevance, and statistical assessment, the final adjusted models included women age, women's education, number of children, pregnancy status, residing status, menstruated status, number of unions, women decision making autonomy, pressure to pregnant, contracept decision, justification of IPV, place of residence, administrative division, religion, wealth, household size, and sex household head. These covariates were adjusted to account for potential confounding in the association between digital connectivity and mental health symptoms.

### 3.5 Model diagnostics and assumptions

Variance inflation factor (VIF) was used to test multicollinearity, and an acceptable level was considered as <5. Model fit was evaluated using both the Pearson chi-square test and the Hosmer–Lemeshow test based on deciles of predicted probabilities where a $p > 0.05$ indicates models' goodness of fit [36].

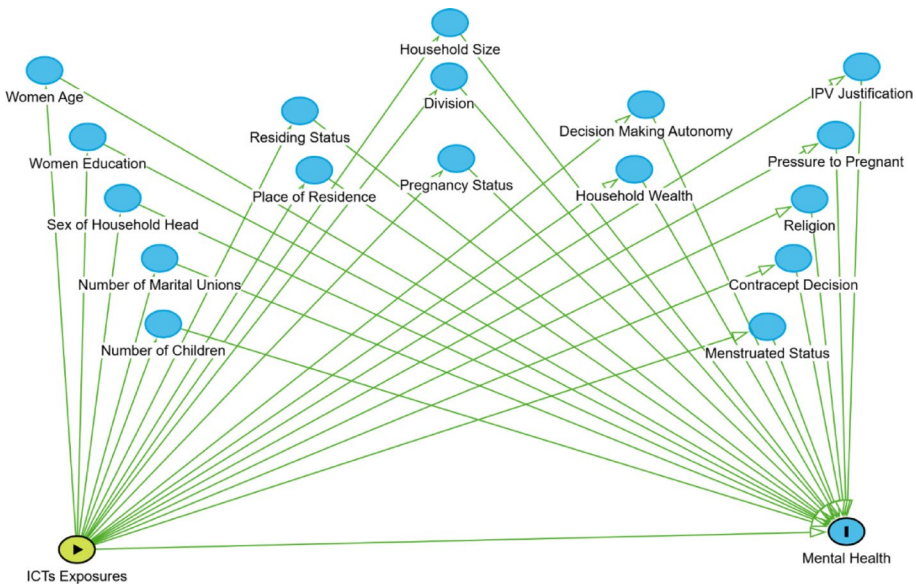



**Fig. 2** DAG showing the conceptual framework of exposure, outcome, and confounders in this study

### 3.6 Statistical analysis

This study conducted descriptive statistics, to summarize the background characteristics of the study variables. Univariates and bivariate (Chi square test) analyses was conducted to assess the initial association. Separate logistic regression models were fitted for each of the exposures (ICT use, ICT ownership, and internet usage) using weights. Each exposure was modeled using three different models: one overall model, one for rural stratification, and one for urban stratification.

Model Specification (for each exposure)

Let $p$ be the probability of anxiety/depression.

1. Overall Model

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1(\text{Exposure}) + \beta_2(\text{Covariates})$$

2. Rural Model

$$\log\left(\frac{p_R}{1-p_R}\right) = \beta_{0R} + \beta_{1R}(\text{Exposure}) + \beta_{2R}(\text{Covariates})$$

3. Urban Model

$$\log\left(\frac{p_U}{1-p_U}\right) = \beta_{0U} + \beta_{1U}(\text{Exposure}) + \beta_{2U}(\text{Covariates})$$

Additionally, spatial analyses were applied to assess administrative division wise geo-spatial inequalities to progress maternal health. All results reported with adjusted odds ratios (AORs) with 95% confidence intervals (CIs) were reported. This study used complete cases analysis.

### 3.7 Analytical tools

For spatial analysis the depression and anxiety prevalence were calculated as division wise percentage by using **STATA 17** exported to MS Excel format and finally, import **R studio 4.5.1**, and using *sf,sp, ggplot2,* and *tmap* packages for visualizations. For complex survey design, sampling weights, with clustering adjustment (primary sampling unit v021) and stratification adjustment (v022) carried out with the ***svyset*** command.

### 3.8 Code availability

All analytical scripts, data of R studio and STATA used for cleaning, preprocessing, statistical analysis are available at this GitHub repository: https://github.com/muhammadsalek/digital-connectivity-mental-health-bangladesh-women/blob/main/README.md.

### 3.9 Ethical considerations

The BDHS 2022 survey protocols were reviewed and approved by the Institutional Review Board (IRB) of ICF International, USA, and the Bangladesh Medical Research Council (BMRC). Written informed consent was given by all participants at the time of initial data collection. For respondents younger than 18 years of age, consent was obtained from the guardian or parent, along with assent from the minor participant. All the data were

de-identified to ensure confidentiality and privacy in accordance with DHS data protection guidelines. However, details of ethical approval for DHS are available at: https://dhsprogram.com/Methodology/Protecting-the-Privacy-of-DHS-Survey-Respondents.cfm.

## 4 Results

### 4.1 Background characteristics

Among the 18,987 women, (4.8%) women reported were depressed, and (4.1%) women reported anxiety symptoms respectively [Supplementary Fig. 1]. The prevalence of anxiety and depression was higher in the women lived in rural compared to the urban women. Out of all women's who were screen positive for anxiety, 66.9% were rural and 33.1% were urban. Similarly, out of all subjects who were screen positive for depression, 67.4% were rural and 32.6% urban. Among all women who showed suffering for anxiety, 66.9% were from rural areas and 33.1% were from urban areas. Similar pattern was revealed for depression (67.4% rural vs. 32.6% urban). These findings indicate the distribution of cases by place of residence and should not be interpreted as inequalities in prevalence between rural and urban women (Fig. 3).

Among women with anxiety, 77.8% revealed internet use and 83.5% owned ICT devices. Additionally, 80.8% of women with depression were internet users, and 81.9% reported ICT ownership (Fig. 4).

Anxiety prevalence associated with age, from 2.1% among women aged 15–19 years to 6.1% among those aged 45–49 years ($\chi^2 = 67.95$, $p < 0.001$), and lower prevalence was observed with higher education (5.2% in primary vs 2.6% in higher education, $\chi^2 = 39.96$, $p < 0.001$). Women with ≥3 children showed higher anxiety (5.6% vs 3.1% among childless women, $\chi^2 = 46.80$, $p < 0.001$), while lower prevalence was observed among women with recent menstruation (3.7% vs 5.1%, $\chi^2 = 20.70$, $p < 0.001$). Women with decision-making autonomy (4.4% vs 3.0%, $\chi^2 = 17.20$, $p < 0.001$), those experiencing pressure to become pregnant (11.4% vs 3.9%, $\chi^2 = 82.07$, $p < 0.001$), those involved in contraceptive decisions (4.2% vs 3.3%, $\chi^2 = 3.95$, $p = 0.047$), and those justifying IPV (4.9% vs 4.0%, $\chi^2 = 4.64$, $p = 0.031$) showed higher anxiety prevalence. Anxiety prevalence varied by division (highest in Khulna 5.6%, lowest in Rajshahi 2.6%, $\chi^2 = 50.55$, $p < 0.001$), religion (Muslim 4.3% vs non-Muslim

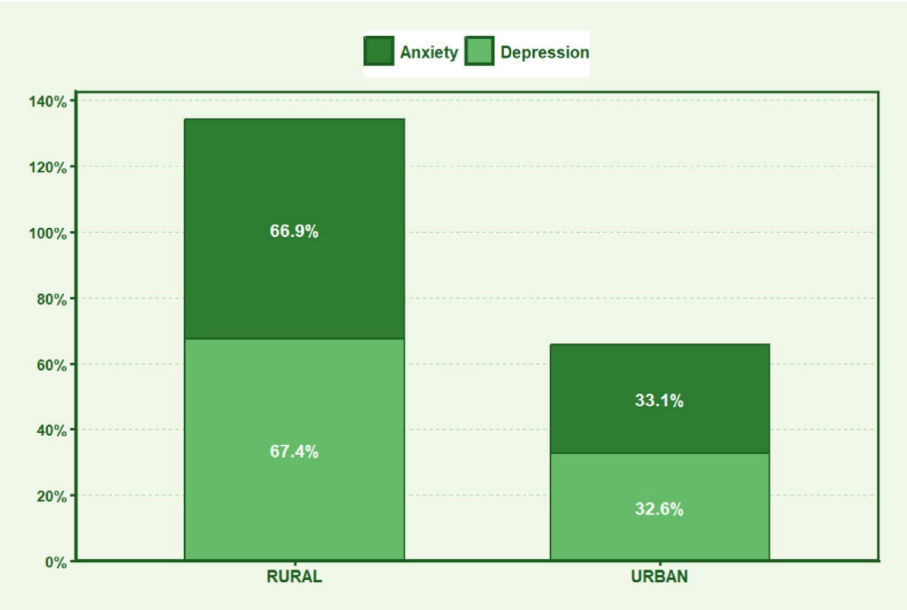



**Fig. 3** Descriptive Inequalities of rural–urban wise mental health symptoms

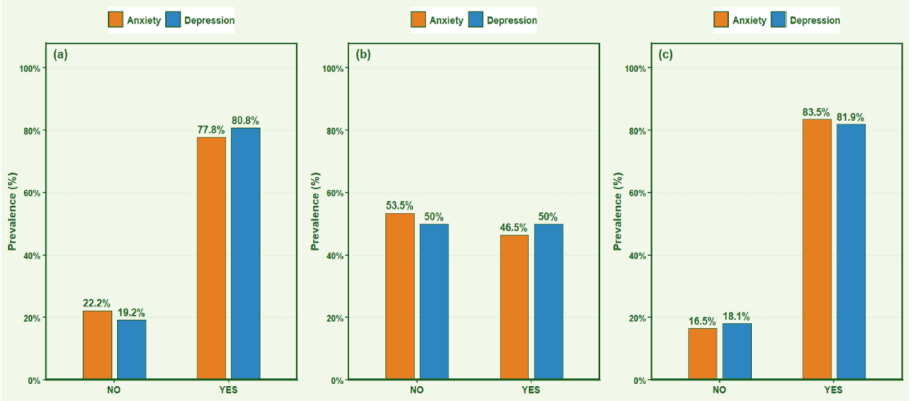



**Fig. 4** Mental health symptoms by various dimension of digital connectivity

2.4%, $\chi^2 = 15.65$, $p < 0.001$), ICT usage (3.8% vs 4.4%, $\chi^2 = 4.04$, $p = 0.045$), and paternal education (5.8% among no education vs 3.0% among secondary complete, $\chi^2 = 40.26$, $p < 0.001$).

Depression prevalence associated with age (3.4% at 15–19 years to 6.9% at 45–49 years, $\chi^2 = 41.33$, $p < 0.001$), and lower prevalence was observed with higher education (5.8% in primary vs 3.5% in higher education, $\chi^2 = 28.96$, $p < 0.001$). Higher parity (≥3 children 5.9% vs 4.4% among childless women, $\chi^2 = 20.99$, $p < 0.001$), multiple unions (7.0% vs 4.7%, $\chi^2 = 6.55$, $p = 0.011$), and decision-making autonomy (5.0% vs 4.2%, $\chi^2 = 5.01$, $p = 0.025$) were associated with higher depression prevalence. Pressure to become pregnant showed a significant difference (12.8% vs 4.6%, $\chi^2 = 83.63$, $p < 0.001$), and preference for fewer children was also associated (6.6% vs 4.7%, $\chi^2 = 4.42$, $p = 0.036$). Regional variation was observed (Rangpur 6.9%, Khulna 6.6%, Rajshahi 3.4%, $\chi^2 = 70.06$, $p < 0.001$), as well as religion (Muslim 5.0% vs non-Muslim 3.5%, $\chi^2 = 9.02$, $p = 0.003$), wealth status (poor 5.2% vs rich 4.3%, $\chi^2 = 9.64$, $p = 0.008$), and paternal education (no education 6.8% vs secondary complete 3.7%, $\chi^2 = 39.39$, $p < 0.001$) (Table 1).

## 4.2 Univariable and bivariable findings

Older age was significantly associated with depression, with OR=1.42 (95% CI 1.01–2.00, $p = 0.044$) for women aged 40–44, and OR=1.91 (95% CI 1.35–2.72, $p < 0.001$) for 45–49, respect to 15–19-year-old women. Higher level of education showed significant association with OR=0.79 (95% CI 0.67–0.93, $p = 0.004$) for incomplete secondary, and 0.52 (95% CI 0.39–0.68, $p < 0.001$) for secondary+ education, with depression. Additionally, having 3+ children was associated with higher odds (OR=1.33, 95% CI 1.00–1.79, $p = 0.052$), while menstrual cycles in the last month was significantly lower odds (OR=0.78, 95% CI 0.67–0.90, $p < 0.001$). Having 2+ unions was associated with higher odds (OR=1.63, 95% CI 1.17–2.27, $p = 0.004$), with highest risk of having been pressured to conceive (OR=3.31, 95% CI 2.43–4.52, $p < 0.001$). Muslim women reported a little higher odd (OR=1.39, 95% CI 1.02–1.90, $p = 0.039$). Higher wealth (rich vs poor) was lower odds (OR=0.80, 95% CI 0.60.99, $p = 0.036$), as was higher paternal education (secondary incomplete OR 0.67, 95% CI 0.54–0.82, $p < 0.001$; secondary+higher: OR=0.51, 95% CI 0.39–0.67, $p < 0.001$). Women in Rangpur showed higher odds compared to Barisal (OR=1.67, 95% CI 1.14–2.44, $p = 0.008$) [Supplementary Table S1]. Unadjusted models showed no statistically significant associations between digital exposures and depression or anxiety [Supplementary Table S2]. In national level, ICT use, highlights a significant association with higher odds of depression (aOR = 1.26, 95% CI 1.05–1.50, $p = 0.009$) only [Supplementary Table S3].

## 4.3 Rural–urban inequality of associations

In the context of rural–urban inequalities, this showed an unequal association between various dimensions of digital connectivity and mental health symptoms. ICT use was significantly associated with higher odds of depression

**Table 1** Distribution of clinically relevant anxiety (GAD-7 ≥10) and depression (PHQ-9 ≥10) according to respondents' characteristics among ever-married women in Bangladesh, BDHS 2022

| Characteristics | Anxiety (GAD-7 ≥10) | | | Depression (PHQ-9 ≥10) | | |
|---|---|---|---|---|---|---|
| | No, n (%) | Yes, n (%) | $\chi^2$ ($p$) | No, n (%) | Yes, n (%) | $\chi^2$ ($p$) |
| *Exposures variables* | | | | | | |
| Internet Use | | | 0.79 (0.929) | | | 1.28 (0.257) |
| No | 1,194 (96.8) | 40 (3.2) | | 1,196 (96.9) | 38 (3.1) | |
| Yes | 4,112 (96.7) | 140 (3.3) | | 4,092 (96.2) | 160 (3.8) | |
| ICT Ownership | | | 0.03 (0.858) | | | 2.38 (0.123) |
| No | 2,967 (95.8) | 129 (4.2) | | 2,930 (94.6) | 166 (5.4) | |
| Yes | 15,240 (95.5) | 651 (4.1) | | 15,142 (95.3) | 749 (4.7) | |
| ICT Usage | | | 4.04 (0.045) | | | 0.001 (0.997) |
| No | 9,065 (95.6) | 417 (4.4) | | 9,025 (95.2) | 457 (4.8) | |
| Yes | 9,142 (96.2) | 363 (3.8) | | 9,047 (95.2) | 458 (4.8) | |
| *Demographic variables* | | | | | | |
| Women's age (years) | | | 67.95 (<0.001) | | | 41.33(<0.001) |
| 15–19 | 1,569 (97.9) | 34 (2.1) | | 1,548 (96.6) | 55 (3.4) | |
| 20–24 | 3,065 (97.4) | 82 (2.6) | | 3,036 (96.5) | 111 (3.5) | |
| 25–29 | 3,317 (96.4) | 123 (3.6) | | 3,286 (95.5) | 154 (4.5) | |
| 30–34 | 3,155 (95.6) | 144 (4.4) | | 3,137 (95.1) | 162 (4.9) | |
| 35–39 | 3,026 (95.0) | 159 (5.0) | | 3,017 (94.7) | 168 (5.3) | |
| 40–44 | 2,232 (95.0) | 119 (5.1) | | 2,221 (94.5) | 130 (5.5) | |
| 45–49 | 1,843 (94.0) | 119 (6.1) | | 1,827 (93.2) | 135 (6.9) | |
| Women's education | | | 39.96 (<0.001) | | | 28.96(<0.001) |
| Below Primary | 6,947 (94.8) | 378 (5.2) | | 6,901 (94.2) | 424 (5.8) | |
| Secondary | 8,481 (96.3) | 327 (3.7) | | 8,416 (95.5) | 392 (4.5) | |
| Secondary+Higher | 2,779 (97.4) | 75 (2.6) | | 2,755 (96.5) | 99 (3.5) | |
| Place of residence | | | 1.16 (0.282) | | | 2.26 (0.133) |
| Urban | 6,364 (96.1) | 258 (3.9) | | 6,324 (95.5) | 298 (4.5) | |
| Rural | 11,843 (95.8) | 258 (3.9) | | 11,748 (95.0) | 617 (5.0) | |
| Administrative division | | | 50.55 (<0.001) | | | 70.06 (<0.001) |
| Barishal | 1,961 (96.5) | 72 (3.5) | | 1,946 (95.7) | 87 (4.3) | |
| Chattogram | 2,687 (95.3) | 134 (4.7) | | 2,707 (96.0) | 114 (4.0) | |
| Dhaka | 2,772 (96.6) | 99 (3.4) | | 2,751 (95.8) | 120 (4.2) | |
| Khulna | 2,339 (94.4) | 139 (5.6) | | 2,315 (93.4) | 163 (6.6) | |
| Mymensingh | 1,990 (97.0) | 61 (3.0) | | 1,980 (96.5) | 71 (3.5) | |
| Rajshahi | 2,376 (97.4) | 63 (2.6) | | 2,356 (96.6) | 83 (3.4) | |
| Rangpur | 2,185 (95.1) | 113 (4.9) | | 2,140 (93.1) | 158 (6.9) | |
| Sylhet | 1,897 (95.0) | 99 (5.0) | | 1,877 (94.0) | 119 (6.0) | |
| Religion | | | 15.65 (<0.001) | | | 9.02 (0.003) |
| Non-Muslim | 1,924 (97.6) | 48 (2.4) | | 1,904 (96.5) | 68 (3.5) | |
| Muslim | 16,283 (95.7) | 732 (4.3) | | 16,168 (95.0) | 847 (5.0) | |
| Wealth category | | | 1.14 (0.565) | | | 9.64 (0.008) |
| Poor | 6,784 (95.8) | 300 (4.2) | | 6,716 (94.8) | 368 (5.2) | |
| Middle Class | 3,625 (95.7) | 161 (4.3) | | 3,585 (94.7) | 201 (5.3) | |
| Rich | 7,798 (96.1) | 319 (3.9) | | 7,771 (95.7) | 346 (4.3) | |
| Household size | | | 0.32 (0.574) | | | 3.26 (0.071) |
| 1–3 members | 3,954 (95.7) | 176 (4.3) | | 3,909 (94.6) | 221 (5.4) | |
| 4 or more | 14,253 (95.9) | 604 (4.1) | | 14,163 (95.3) | 694 (4.7) | |
| Household head sex | | | 0.02 (0.876) | | | 0.53 (0.469) |
| Male | 16,000 (95.9) | 684 (4.1) | | 15,873 (95.1) | 811 (4.9) | |
| Female | 2,207 (95.8) | 96 (4.2) | | 2,199 (95.5) | 104 (4.5) | |
| Currently working | | | 2.20 (0.138) | | | 0.27 (0.604) |
| No | 12,754 (96.0) | 527 (4.0) | | 12,648 (95.2) | 633 (4.8) | |
| Yes | 5,453 (95.6) | 253 (4.4) | | 5,424 (95.1) | 282 (4.9) | |
| Husband education | | | 40.26 (<0.001) | | | 39.39 (<0.001) |

**Table 1** (continued)

| Characteristics | Anxiety (GAD-7 ≥10) | | | Depression (PHQ-9 ≥10) | | |
|---|---|---|---|---|---|---|
| | No, n (%) | Yes, n (%) | $\chi^2$ ($p$) | No, n (%) | Yes, n (%) | $\chi^2$ ($p$) |
| No education | 2,293 (94.2) | 140 (5.8) | | 2,267 (93.2) | 166 (6.8) | |
| Primary incomplete | 2,269 (95.0) | 119 (5.0) | | 2,273 (95.2) | 115 (4.8) | |
| Primary complete | 2,448 (95.3) | 120 (4.7) | | 2,424 (94.4) | 144 (5.6) | |
| Secondary incomplete | 6,371 (96.2) | 251 (3.8) | | 6,315 (95.4) | 307 (4.6) | |
| Secondary complete | 4,826 (97.0) | 150 (3.0) | | 4,793 (96.3) | 183 (3.7) | |
| *Maternal variables* | | | | | | |
| Number of children | | | 46.80 (<0.001) | | | 20.99 (<0.001) |
| None | 1,944 (96.9) | 62 (3.1) | | 1,918 (95.6) | 88 (4.4) | |
| 1–2 | 10,801 (96.5) | 395 (3.5) | | 10,710 (95.7) | 486 (4.3) | |
| ≥3 | 5,462 (94.4) | 323 (5.6) | | 5,444 (94.1) | 341 (5.9) | |
| Menstruated last 6 weeks | | | 20.70 (<0.001) | | | 15.39 (<0.001) |
| No | 5,470 (94.9) | 294 (5.1) | | 5,433 (94.3) | 331 (5.7) | |
| Yes | 12,737 (96.3) | 486 (3.7) | | 12,639 (95.6) | 584 (4.4) | |
| Currently abstaining | | | 0.07 (0.786) | | | 0.07 (0.790) |
| No | 17,519 (95.9) | 752 (4.1) | | 17,392 (95.2) | 879 (4.8) | |
| Yes | 688 (96.1) | 28 (3.9) | | 680 (95.0) | 36 (5.0) | |
| Number of Unions | | | 1.51 (0.218) | | | 6.55 (0.011) |
| Once | 17,607 (95.9) | 748 (4.1) | | 17,484 (95.3) | 871 (4.7) | |
| More than once | 600 (94.9) | 32 (5.1) | | 588 (93.0) | 44 (7.0) | |
| Fertility preference | | | 2.73 (0.098) | | | 4.42 (0.036) |
| Fewer | 563 (94.8) | 31 (5.2) | | 555 (93.4) | 39 (6.6) | |
| Same/More | 16,229 (96.1) | 655 (3.9) | | 16,091 (95.3) | 793 (4.7) | |
| Missing | 1,415 (93.8) | 94 (6.2) | | 1,426 (94.5) | 83 (5.5) | |
| *Reproductive variables* | | | | | | |
| Pressure to become pregnant | | | 82.07 (<0.001) | | | 83.63 (<0.001) |
| No | 17,687 (96.1) | 713 (3.9) | | 17,560 (95.4) | 840 (4.6) | |
| Yes | 520 (88.6) | 67 (11.4) | | 512 (87.2) | 75 (12.8) | |
| Contraceptive decision | | | 3.95 (0.047) | | | 0.34 (0.560) |
| Husband/Other | 1,871 (96.7) | 63 (3.3) | | 1,846 (95.4) | 88 (4.6) | |
| Respondent | 16,336 (95.8) | 717 (4.2) | | 16,226 (95.1) | 827 (4.9) | |
| *Contextual variables* | | | | | | |
| Decision Making Autonomy | | | 17.20 (<0.001) | | | 5.01 (0.025) |
| No | 4,347 (97.0) | 136 (3.0) | | 4,295 (95.8) | 188 (4.2) | |
| Yes | 13,860 (95.6) | 644 (4.4) | | 13,777 (95.0) | 727 (5.0) | |
| Health card access | | | 0.08 (0.782) | | | 0.00 (0.947) |
| No | 17,952 (95.9) | 770 (4.1) | | 17,820 (95.2) | 902 (4.8) | |
| Yes | 255 (96.2) | 10 (3.8) | | 252 (95.1) | 13 (4.9) | |
| Justification of IPV | | | 4.64 (0.031) | | | 2.98 (0.084) |
| No | 15,843 (96.0) | 658 (4.0) | | 15,723 (95.3) | 778 (4.7) | |
| Yes | 2,364 (95.1) | 122 (4.9) | | 2,349 (94.5) | 137 (5.5) | |

symptoms among rural women (aOR=1.38, 95% CI 1.12–1.69; $p < 0.001$) only. Surprisingly, ICT ownership associated with higher odds of depression symptoms (aOR=1.81, 95% CI 1.01–3.25; $p = 0.047$), anxiety symptoms (aOR=2.11, 95% CI 1.14–3.92; $p = 0.018$) among urban women only (Table 2).

### 4.4 Descriptive spatial inequalities of depression symptoms associated to digital connectivity

Depression prevalence showed notable geographic variation and generally consistent patterns of digital connectivity. Chattogram, Sylhet, Dhaka, and Barishal revealed high depression prevalence along with higher access across at least two digital connectivity domains. Khulna, Rajshahi, and Mymensingh revealed moderate levels of both digital connectivity and depression symptoms, while Rangpur reported the lowest digital connectivity and

**Table 2** Rural- urban inequalities of digital connectivity and mental health symptoms

| Exposure | Rural aOR (95% CI) | *p*-value | Urban aOR (95% CI) | *p*-value |
|---|---|---|---|---|
| *Depression symptoms* | | | | |
| Internet use | | | | |
| Yes | 1.37 (0.73–2.56) | 0.321 | 1.13 (0.49–2.66) | 0.77 |
| No (Reference) | 1.00 (Reference) | – | 1.00 (Reference) | – |
| ICT usage | | | | |
| Yes | 1.38 (1.12–1.69) | <0.001 | 0.97 (0.69–1.35) | 0.87 |
| No (Reference) | 1.00 (Reference) | – | 1.00 (Reference) | – |
| ICT ownership | | | | |
| Yes | 0.98 (0.79–1.22) | 0.89 | 1.81 (1.01–3.25) | 0.047 |
| No (Reference) | 1.00 (Reference) | – | 1.00 (Reference) | – |
| *Anxiety symptoms* | | | | |
| Internet use | | | | |
| Yes | 1.22 (0.68–2.19) | 0.497 | 1.85 (0.83–4.15) | 0.132 |
| No (Reference) | 1.00 (Reference) | – | 1.00 (Reference) | – |
| ICT usage | | | | |
| Use | 1.03 (0.83–1.29) | 0.749 | 0.93 (0.66–1.30) | 0.675 |
| No (Reference) | 1.00 (Reference) | – | 1.00 (Reference) | – |
| ICT ownership | | | | |
| Yes | 1.04 (0.81–1.33) | 0.744 | 2.11 (1.14–3.92) | 0.018 |
| No (Reference) | 1.00 (Reference) | – | 1.00 (Reference) | – |

Adjusted Odds ratios (aORs) with 95% confidence intervals (CIs) were estimated using multivariable logistic regression models, stratified by rural and urban residence. All models account for the complex survey design. Models were fully adjusted for women's age, education, number of children, pregnancy status, residence status, menstrual status, union status, empowerment status, reproductive pressure, contraceptive decision-making autonomy, justification of intimate partner violence, administrative division, religion, household wealth, household size, and sex of household head

the lowest prevalence of depression. These descriptive findings reflect spatial heterogeneity but do not imply a causal association (Fig. 5).

### 4.5 Descriptive spatial inequalities of anxiety symptoms associated to digital connectivity

High digital connectivity was showed in Chattogram, Sylhet, Dhaka, and Barishal, considered by higher values across at least two digital connectivity domains (internet and ICT use) and higher anxiety symptoms. Khulna and Rajshahi revealed moderate digital connectivity and medium anxiety symptoms. While, Rangpur reported low digital connectivity across all domains of digital connectivity and the lowest anxiety symptoms, these findings should be interpretated as descriptive only (Fig. 6).

## 5 Discussion

This study examines the association between digital connectivity and mental health symptoms among women in reproductive aged through rural–urban, and administrative division-wise descriptive disparities using nationally representative cross-sectional data from BDHS 2022. Overall, around 4.8% of women reported experienced depression symptoms and 4.1% reported experienced anxiety symptoms.

Given the cross-sectional design, causal relationships cannot be established, and reverse causation remains possible, as mental health symptoms may also associates with digital connectivity. A substantial proportion of women experiencing anxiety and depression resided in rural areas, highlighting persistent geographic disparities in mental health symptoms. Initially, many women with anxiety and depression reported internet use and ICT ownership, only ICT use remained significantly associated with depression in the overall analysis.

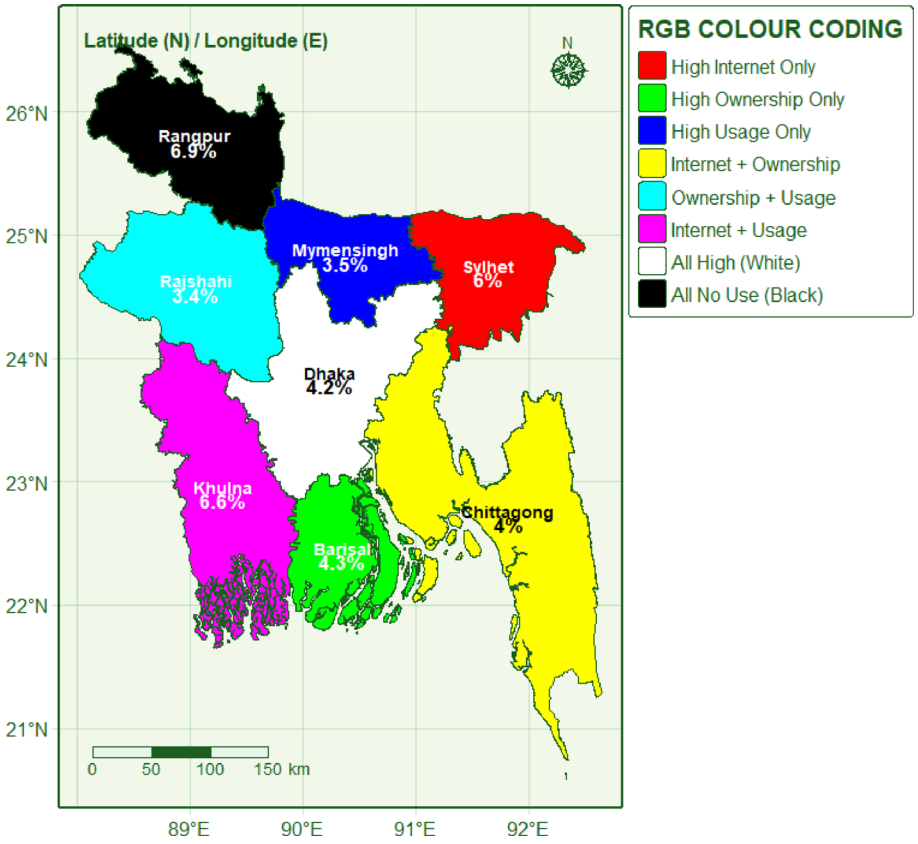



**Fig. 5** Spatial inequalities of digital connectivity and depression symptoms

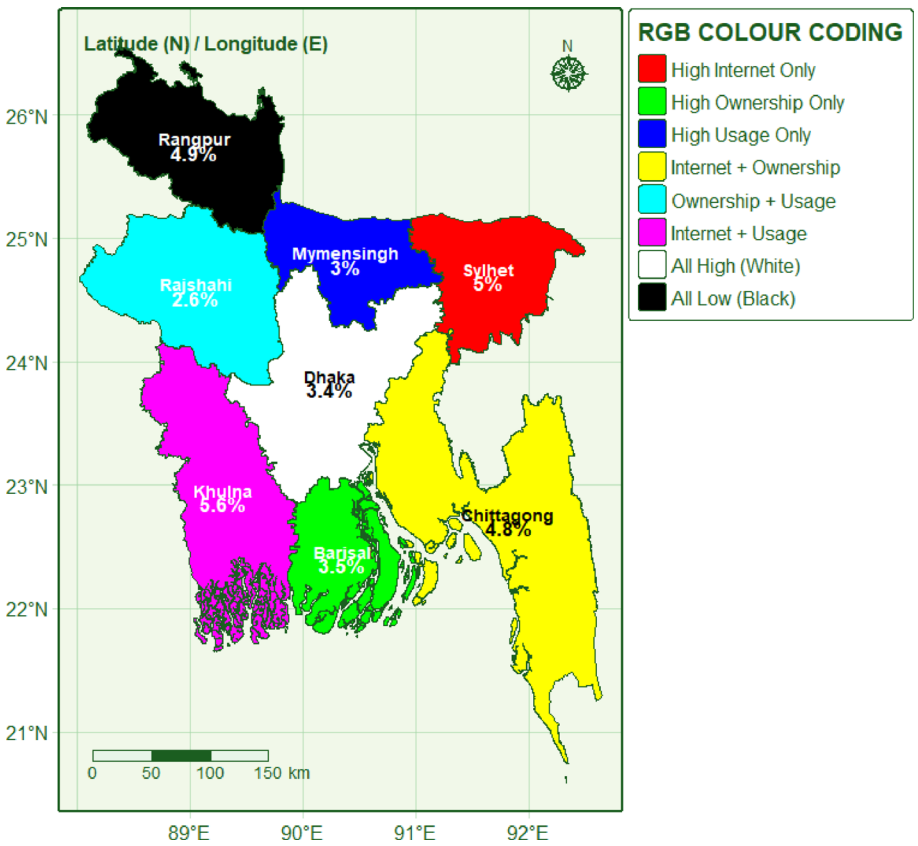



**Fig. 6** Spatial inequality of digital connectivity and anxiety symptoms in Bangladesh

The stratified analyses further suggested contextual differences, with ICT use associated with depression among rural women, whereas ICT ownership was associated with both anxiety and depression among urban women. Regional variation was also evident, as divisions with higher digital access, including Dhaka, Chattogram, Sylhet, and Barishal, associates with higher burden of mental health symptoms, while Rangpur showed comparatively lower digital access and lower prevalence. These findings suggest that the relationship between digital connectivity and mental health may vary across geographic and social contexts. Strengthening equitable digital inclusion alongside accessible, context-specific mental health services may help reduce existing disparities, particularly among women living in underserved areas [41].

These findings may indicate basic social and structural disparities, indicating that mental health symptoms disparities are analytically associated with administrative division and various components of digital connectivity [42]. Due to inadequate socioeconomic and mental health resources and supports, women residing in rural residence may be associated with higher anxiety symptoms and depression symptoms [43]. The observed association between ICT use and depression among rural women may showed that higher digital access, in the limited of optimum digital connectivity, mental health symptoms awareness, and support services, associated with poorer mental health symptoms. While, given the cross-sectional design, causal inferences cannot be established [44].

In urban contexts, ICT ownership was significantly associated with both anxiety and depression symptoms. Urban women may experience higher levels of information overload, expectation to seek higher amount of ICT ownership, social comparison, work–family conflict, and exposure to difficult online content, all of which may coldly associated with mental health symptoms [45]. These findings indicates that digital connectivity alone may be inadequate to improve mental health symptoms and, in the limited supportive resources, may be associated with higher mental health symptoms [46, 47]. Generally, mental health symptoms were more common in areas with higher digital connectivity, where higher online access was associated with a higher burden of anxiety and depression [41].

While, regions with limited access of digital connectivity, including Rangpur, observed lower prevalences of anxiety and depression symptoms. Conversely, this should not be interpreted as a positive association of digital exclusion; rather, it indicates that growing digital connectivity without equitable mental health support may intensify existing inequalities [48].

Previous studies also reported, older women, women with lower education, higher parity and economic inequalities associated with mental health symptoms. Additional study showed, internet use is significantly associated with mental health symptoms in Bangladesh [16]. Our results consistent these trends and add to the literature by including the digital connectivity as a significant dimension of equity [49]. Previous research in Bangladesh has centered mostly on socioeconomic and reproductive factors on women mental health and limited explored regard to digital connectivity [50]. Although digital connectivity revealed to help the distribution of mental health information, our findings consistent with this substantial evidence suggesting that it may also be associated with poorer mental health symptoms in LMICs [51, 52]. The significant association between mental health symptoms and pressure to become pregnant as identified in this study is not new as previous qualitative and quantitative studies have outlined reproductive coercion as one of the key mental health symptom's risk factors among Bangladeshi women [16].

Overall, this study explores a topic that remains limited explored in Bangladeshthe existing literature by showing that the association between digital connectivity and mental health symptoms differs across rural–urban settings and geographic regions [53].

## 6 Conclusions

This study revealed that rural–urban and regional disparities and the association between digital connectivity and mental health symptoms among women in Bangladesh. Although internet use, ICT usage, and ownership are significantly associated with anxiety and depression indicating, notable differences, across Bangladesh. ICT use

was associated with depression among rural women, while ICT ownership was associated with both anxiety and depression among urban women, providing that the association between digital connectivity and mental health symptoms differs across contexts.

These findings reveal the importance of addressing structural disparities, including unequal digital access and limited availability of mental health services. Integrating digital inclusion with equitable, geographically targeted mental health interventions may help reduce these disparities and support progress toward Sustainable Development Goal 3 (Good Health and Well-being).

### 6.1 Recommendations on policies and suggestions

These findings suggest an important policy implication for achieving SDGs 3. Policies should be prioritized equitable digital access along with mental health symptoms literacy and mental support for women, while considering the distinct needs of rural and urban populations. Integrating mental health elevation into digital health and community-based programs may help reduce existing disparities. The observed regional disparities also offer evidence to guide geographically targeted interventions and future research.

### 6.2 Strengths

This study uses nationally representative BDHS 2022 data and validated mental health screening tools (GAD-7 and PHQ-9) to examine associations between digital access and mental health outcomes. It further applies rural–urban stratified and geospatial analyses to capture regional variation and contextual differences in Bangladesh. The inclusion of a broad set of sociodemographic, reproductive, and empowerment-related covariates strengthens the robustness and internal validity of the findings.

### 6.3 Limitations

The cross-sectional study design limits causal inference, limited detail on ICT use, and the potential for reverse causality ie, that digital behavior associated with poor mental health remains. The internet-use analysis included a smaller analytical sample (n = 5,486) due to complete-case analysis among eligible participants is additional limitation. Although no imputation was required, this sample reduction may introduce selection bias and limit the generalizability and statistical power of the findings associates with internet use.

Additionally, ICT use and internet exposure were self-reported and inadequate to available BDHS procedures, without comprehensive evidence on frequency, quality, or purpose of digital engagement. Therefore, dimension limits and measureless factors, such as IPV, online content quality, and social sustenance. However, that this study did not conduct any formal statistical interaction analyses to assess whether rural–urban residence modifies the association between digital connectivity and mental health symptoms.

Therefore, rural–urban differences observed in stratified analysis should be interpretated as descriptive rather than evidence of statistical effect modification. Furthermore, this study did not conduct any formal spatial autocorrelation (Moran's I), or spatial weighted regressions, as the results of the spatial analysis can only be interpretated as descriptive.

### 6.4 Future research directions

Longitudinal methods must be employed in future investigations to help better understand the dynamics and possible causality of associations between digital connectivity and mental well-being in women. There is a need for more refined measurement of digital connectivity that encompasses duration on electronic devices, types of content consumed, and interactions within the digital space in order to understand the mechanisms connecting ICT and mental well-being. Qualitative and hybrid methods could be considered in order to provide more information

on women's experience with exposure to digital media. The success of equity-based digital mental health interventions must also be investigated. Overall, future study should be conducted advanced spatial techniques to understand the spatial pattern of determinants.

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**Author contributions** Md Salek Miah: Conceptualization, Data Curation, Methodology, Formal Analysis, Writing—Original Draft, Resources, Project Administration, and Finalized the Manuscript. Md Jamal Uddin (Corresponding Author): Review, Editing, Methodology Developments, Resources, Supervision, and Finalized the Manuscript.

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**Data availability** The data used in this study are publicly available from the Bangladesh Demographic and Health Survey (BDHS) 2022. Access to the dataset can be requested through the DHS Program website (https://dhsprogram.com/) following their data access policies.

**Code availability** The code materials of this study are available from the corresponding author upon reasonable request.

## Declarations

**Consent for publication** Not applicable.

**Clinical trial number** Not applicable.

**Competing interests** The authors have no interest.

**Declaration of generative** The authors used AI tools (Grammarly, QuillBot, ChatGPT GPT-4) solely to improve language and clarity. No AI was used for data generation, analysis, or interpretation. The authors reviewed all content and confirm the manuscript's accuracy, originality, and integrity.

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